# A Study of the Mathematics of the Mayan Empire

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**Abstract:**

This thesis will discuss the mathematics that has developed from the study of Mayan culture. Archeological evidence show that the Mayas developed a numerical number system used for counting and quantifying information and a calendar system used to record dates over long periods. However, due to a lack of empirical evidence, no one knows for sure how the Mayas actually did calculations, but mathematical ideas developed from what has been found and thus lead us to possible ways of solving the mathematical problems that arise from the number system and calendar system of the Mayas. The study of the numerical and calendar system of the Mayas concludes that contrary to what the Spaniards thought during the age of imperialism, there is considerable evidence showing the Mayas were indeed quite an advanced group of people. Even though no one knows how the Mayas solved problems, we do know that the Mayas utilized mathematical systems that required a sophisticated degree of mathematical knowledge.

**Brief History of the Mayas**

Contrary to the popular belief held by the Europeans nearly five hundred years ago, the Mayas had one of the world's most advanced ancient civilizations, comparable to Mesopotamia, ancient Egypt, ancient India, and ancient China until the Spanish came and destroyed the culture they had created. By the time the famous Spanish conquistador Hernán Cortés conquered the Mayas, they had already devised and used a complex calendar system that demonstrated their mathematical and scientific sophistication. In addition, they developed and used a base twenty place-value number system that included a zero that showed they understood the importance of zero in a place value system.

It is important before discussing the mathematical and scientific ideas of the Mayas that we become acquainted with the Mayas in order to put their work in a context. The Mayas were one of the major groups of people that developed a large civilization in Mesoamerica before the Spanish conquest in the sixteenth century. They occupied parts of southern Mexico, including the modern Mexican states of Tabasco, Chíapas, Campeche, Yucatán, and Quintana Roo. They also occupied parts of Central America including the modern day countries of Belize, Guatemala, El Salvador, and Honduras. Today there are descendents of the Mayas that live in Guatemala and southern Mexico.

The Mayas themselves are descendents of the people who migrated from Asia during the *Paleo-Indian* period, which took place during the last Ice Age, from about 20, 000 B.C. to 8,000 B.C. Once they settled in Mesoamerica, they became mostly hunter and gatherers until a new period began at about 8,000 B.C. known as the *Archaic* period. This period notes the transitional period from a life of hunting and gathering to a life of living in small villages where people farm and live in houses on a permanent basis. During this transition, the population started to increase due to the increased availability of food.

At about 2,000 B.C., the *Pre-Classic* period began. During this period, the first Mayan civilizations consisting of established villages and towns appeared. Large amounts of pottery and temple mounds developed during this period, and there was a population explosion due to improved farming techniques. The complexity of the form of the pottery demonstrated the increased civilization of Mesoamerica, for according to Encarta.msn.com, "the earliest pottery was very complex both in forms and decoration. It seems to have had a social function and been used primarily for ritual feasting.".[8]

In addition, the building of these temple mounds as well as their size, location, and design indicate that ceremonial rituals were practiced in Mesoamerica at that time. For "one large mound with a clay foundation ….is obviously a larger form of the private homes in the area. The size and location suggest it was a used as a temple where public rituals were carried out. These temple mounds also hint at an organized priesthood supported by the group".[13]

In addition, a clear distinction between the elite class and the commoners developed. The elite class was a small ruling class consisting of priests and chiefs. Most people were commoners and they worked the land, did the labor, and supported the ruling class under the direction of the elite. The elite class enjoyed a higher standard of living than did the commoners, which also shows the development of a socioeconomic system and advancement towards a civilization. According to Encarta.msn.com,

> The villages of chiefs were greater in size and importance than other settlements, possibly housing more than 1000 residents. These villages became centers of political and economic activity. They were also important religious centers, boasting more elaborate ceremonial sites—such as temple mounds and sacred enclosures—than other villages. From these chiefly villages, Mesoamericans began to form the complex societies that would come to define the region.[8]

There have been trade routes discovered near the Mexican border to Guatemala, which showed that a system of trade also developed during the Pre-Classic era. Later during this era, the first states and a writing system developed from these civilizations, but towards the end of this period (250 A.D.), they experienced a decline. However, they revived and further developed into one of the most powerful collections of states in Mesoamerica, known as the Mayan empire.

The *Classic period*, which started in 250 A. D. was marked by the continued growth of states, which lead to the rise of the Mayan empire, which peaked during this era. Large population centers including big cities developed. For, "cultural centers experienced great population increases reaching 30,000 to 40,000 inhabitants."[11] The most well known are the cities of Teotihuacan and Monte Alban. Teotihuacan is near present-day Mexico City and Monte Alban is in the present day Mexican state of Oaxaca.

Large political systems with head rulers also developed during this period. Some of the political systems were "divine kingships" ruled by kings. Under the divine kingships, "kings claimed to descend from gods and often orchestrated their appearances to reinforce this idea".[11] Three important dynasties took place. One well known divine kingship was the rein of Pakal who "was the greatest Mayan ruler of Palenque, Chíapas…" and also known as "The Sacred Lord of Palenque". [11] Another important divine kingship was the Jaguar Dynasty of which "the most noteworthy kings of Yaxchilán, Chíapas were Escudo Jaguar(Jaguar Shield) and Pájaro Jaguar(Jaguar Bird), a father and son combination which appeared on numerous stelae, signifying power and victory". The third important kingship was the rein of 18 Rabbit (18 Conejo) "who was the thirteenth ruler of Copán in Honduras…credited with transforming the Copán art of sculpture." [11] The Mayan empire had a similar system of rule, as did the ancient Chinese, who also had a powerful empire ruled under powerful dynasties.

The Classic period was also a time and period of rapid artistic, scientific, and political development which created the distinction of the Mayan empire from other civilizations in the western hemisphere. For, "during this period of accelerated changes and development, the Mayas became deeply involved in research in the fields of astronomy and mathematics." [11] It was

during this era the Mayas developed the vigesimal number system, the zero, and complex calendar system we will discuss. In addition, they performed accurate astronomical calculations, predictions, and discoveries.

This era also included the use of a complex economic system in which many people were trading merchants and they used beans called cacao as currency. During the Classic era, there was rapid sophistication of their writing systems and the creation of their cosmology system which explained the "creation of the universe, the birth of human kind, their life on earth and their hope for admission to the world beyond an underworld with similarities to Christianity's heaven and hell". [11] Towards the end of the Classic period, which ended in 900 A.D., Classic Maya experienced a decline.

By 900 A. D., with the exception of the Yucatán Peninsula, Classic Maya collapsed, and the jungles buried the Classic cities in those regions for thousands of years. At this point, the *Post Classic* period began, which terminated when the Spanish conquistador Hernán Cortés conquered Post-Classic Maya. The Post-Classic culture was different from the classic culture with respect to the political system and focus of the culture, for it was more oriented towards warfare, more secular, and less artistic. This change all started when a group of sailors known as Itzá arrived in the Yucatan peninsula. As soon as the Itzá arrived, they successfully gained control over the Mayan states in the Yucatan peninsula and the elite Itzá ruled the Post-Classic Maya.

Under the Itzá, Maya experienced rule under joint rule instead of divine kingship as in the Classic Period. They converted the Yucatan peninsula into "a conquest state with commercial interests" and did so by manipulation of "diverse Maya and Mexican political and artistic

expressions to reinforce and consolidate their authority, and rationalize their expansionist policies" [12]. The Itzá incorporated into the Post-Classic Mayan culture elements of their own culture including the "cult of kukulcán, (or Quetzalcóatl, the plumed serpent of the Toltec, and filled the city with its iconography: skulls, eagles, and jaguars" (SITE SOURCE Fuentes).

Post Classic Maya also lacked artistic architectural development, for the Post Classic Mayas simplistically made the buildings with no particular décor or design. According to Fuentes, "the architecture of Post Classic settlements other than Chichén Itzá, the capital city of Post Classic Maya, was simple, basic, even crude characterized by low, square, or rectangular rooms, no vaulting, and a lack of decorative detail." Buildings were built primarily for the purpose of occupation, not for ceremonial matters, which is to the contrary of the Maya Classic era.

Post Classic Maya continued to flourish in this warfare state until the decline of Chichén Itzá due to warfare. Then the activity shifted to another city called Mayapán, which also eventually fell to warfare. Afterwards activity shifted again, the peninsula divided into several states, and Post Classic Maya became more focused on trade. During this period, many communities earned a living by trading goods with others. Some people traded bees, others traded fish, and others traded crops. Post Classic Maya remained a trade oriented society until the era of the Spanish conquest.

**The Number System of the Mayas**

The Mayas developed their number system for the purpose of counting such as according to Fuentes, "objects, people, distances, and even cacao seeds".(Fuentes Conversación y Gramática). The numerical system was especially important in the economy of the Mayas, for

the Mayas used cacao as currency and therefore needed a way of counting the amount of cacao they had, the amount of cacao they had to pay to obtain a god or service, and to mark prices. Since there is no evidence that the Mayas used fractions, irrationals, or negative numbers, the Mayan numerals consists of exclusively the non-negative whole numbers.

The Mayan number system was a vigesimal (base 20) place value number system that used only three symbols to represent numbers which were •, a dot representing one unit, ▬, a bar representing five units, and , a shell representing zero units. The following table from the website **http://www.tacomacc.edu/HOME/jkellerm/MATH106/EthnomathematicsText/Chapter2/MayanNumerals.htm** demonstrates how to represent in Mayan form numbers 0 to 19 using these symbols.

| | | | | |
|---|---|---|---|---|
| 0 | • 1 | • • 2 | • • • 3 | • • • • 4 |
| ▬ 5 | • ▬ 6 | • • ▬ 7 | • • • ▬ 8 | • • • • ▬ 9 |
| ▬ ▬ 10 | • ▬ ▬ 11 | • • ▬ ▬ 12 | • • • ▬ ▬ 13 | • • • • ▬ ▬ 14 |
| ▬ ▬ ▬ 15 | • ▬ ▬ ▬ 16 | • • ▬ ▬ ▬ 17 | • • • ▬ ▬ ▬ 18 | • • • • ▬ ▬ ▬ 19 |

**Table. 1**

Since the Mayan number system is a vigesimal number system, it is a number system consisting of place values called the ones( $20^0$), the twenties ($20^1$), the four hundreds ( $20^2$), the eight thousands ( $20^3$), the one hundred sixty thousands ($20^4$), etc. The Mayas wrote the place values in the form demonstrated below:

One hundred Sixty Thousands

Eight Thousands

Four Hundreds

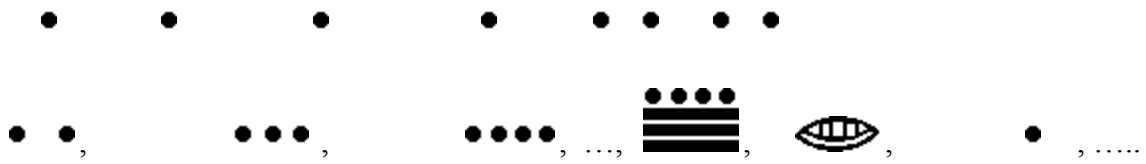
Twenties

Ones

We can think of these place values as *levels*. The Mayas wrote the numbers using this format. Since the Mayan number system is vigesimal, they placed exclusively the Mayan representation of numbers 0 to 19 as described in table 1 in the levels to represent numbers. Every symbol in the ones place value represents the same number of units as previously discussed. Every symbol in the twenties place value represent twenty times more units than every equivalent symbol in the ones place value. Likewise, every symbol in the four hundreds place value represent twenty times more units than every equivalent symbol in the twenties place value, every symbol in the eight thousands place value represent twenty times more units than every equivalent symbol in the four hundreds place value, etc. The sum of the number of units each placed symbol represents tells us what number we are representing. Therefore, the Mayas wrote the number 20 as described below:

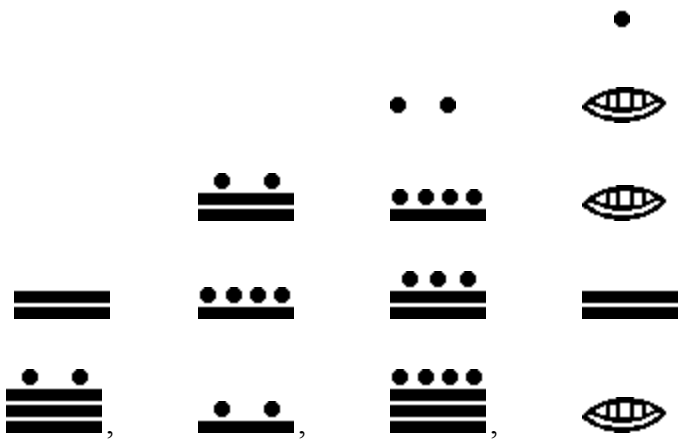
because one dot in the twenties place value represents twenty units and a zero in the ones place value represents zero units which give us a total number of 20 units. Likewise, the Mayas wrote the number 25 as described below:

because a dot in the twenties place value represents 20 units and a dot in the ones place value represents 1 units which give us a total number of 21 units. Likewise the Mayas wrote the numbers 22, 23, 24,…., 39, 40, 41,…. as



respectively.

Likewise, the Mayas wrote the numbers 217, 4987, 19,879, and 160, 200 as :



respectively.

To make the process of conversion from Hindu-Arabic decimal numbers to Mayan numbers more understandable to a beginner, I developed an algorithm for writing the Mayan representation of a decimal number. Let N be the decimal number you want to convert to Mayan form and L be the number of levels N will have in Mayan form. To find L, we first have to find m such that $20^m = N$. To find m, we can apply the natural logarithm to both sides of the equation to get $m = \ln(N)/\ln(20)$. Then to take into account the existence of the first level, let $L = [m] + 1$

where [m] is the greatest integer less than or equal to m. So, if we were to convert 100 to Mayan form, we would first have to find out how many levels (L) 105 have in Mayan form. We can find that out by finding an m such that $20^m = 100$. To solve this equation we apply the natural logarithm to both sides of the equation to get

m = 1.5535… Then we have [m] = 1 which means there are 1 +1 = 2 levels in the Mayan representation of 105. Similarly, if we were to write 2105 in Mayan form, we find out the number of levels in Mayan form by finding m such that $20^m = 2105$. Applying the natural logarithm to both sides, we have m = 2.5543….and [m] = 2 which means there are 2 +1 = 3 levels in the Mayan representation of 2105.

Now, the next step in the algorithm is to find out what symbols and how many of each to place in each level. To do this, consider the top level first and work your way down. Divide N by $20^{L-1}$ according to the Division Algorithm writing N as $N = 20^{L-1}q + r$ where r is the remainder upon division. Place the Mayan representation of q into this level. Repeat this procedure for r, dividing r by $20^{L-2}$ according to the Division Algorithm writing r as q′ where $r = 20^{L-2}q' + r'$. Place the Mayan representation of q′ in the next level down (the second to topmost level). Continue in like fashion until you determine the Mayan numeral to place in the bottom level and then write that number in Mayan form in the bottommost level.

So if we were to finish representing 105 in Mayan form as discussed in the last paragraph, we would apply the Division Algorithm to write 105 as $20^1(5) + 5$. We would then write 5 in Mayan form (▬) in the top level (the second level) and apply the division algorithm again for 5, writing 5 as

$20^0(5) + 0$. Then we would write 5 in Mayan form (▬) in the next level down (the bottom level). Since we have already written all the Mayan symbols in the bottom level, we are done. The final answer is

▬

▬.

Similarly, if we were to finishing representing 2105 in Mayan form, we would apply the Division Algorithm to 2105, writing 2105 as $2105 = 20^2(5) + 105$. Then we write 5 in Mayan form (▬) in the top level. Then we apply the Division Algorithm to 105, writing 105 as $105 = 20^1(5) + 5$ and so we write 5 in Mayan form (▬)in the next level down ( the second level). Finally we apply the Division Algorithm to 5, writing 5 as $5 = 20^0(5) + 0$. We write 5 in Mayan form (▬) in the next level down (the bottom level). Now that we have covered all levels, we are done and so our final answer is

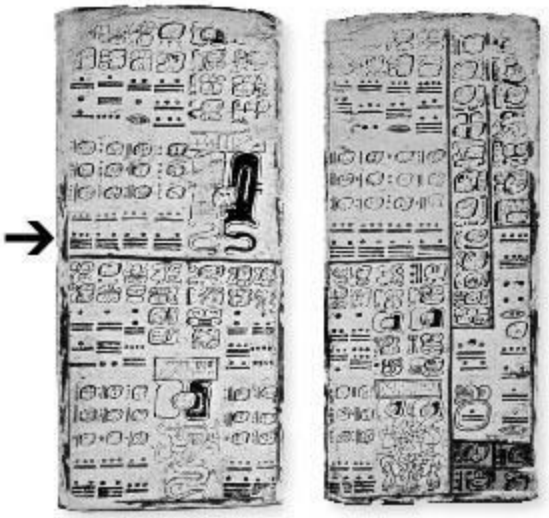
▬

▬

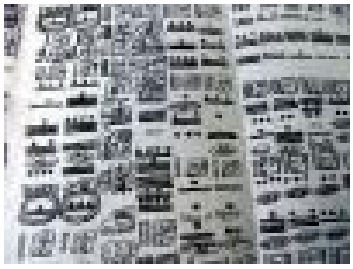
▬.

Below are a few pictures from the websites http://www.hermetic.ch/cal_stud/maya/cd5024.jpg, http://www.astrosociety.org/education/publications/tnl/31/images/fig12.jpg, and http://thumb2.webshots.net/t/59/159/8/14/13/2230814130045485662ZxnDOo_th.jpg respectively consisting of Mayan writing of numerals on stone walls.

**Picture 1.** Writing of Mayan numerals and glyphs on a wall.



**Picture 2.** Mayan writing on a wall including Mayan numerals and glyphs.



**Picture 3.** Mayan writing of numbers in a book.

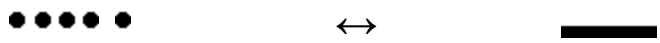
The zero turned out to be an essential part of the Mayan number system given that it is a vigesimal system. The Mayas realized that if there was no zero in their system they would have no way to differentiate the Mayan representations of 1 from 20, 2 from 40, 3 from 60, 4 from 80, 5 from 100, 6 from 120, etc since only the Mayan representation of numbers 1 to 19 can fit in any particular level. Therefore it would have been impossible for the Mayas to represent numbers greater than 19. Therefore, they created the zero to function as a placeholder indicating the absence of units (zero units) in a particular level so that they can represent an infinite amount

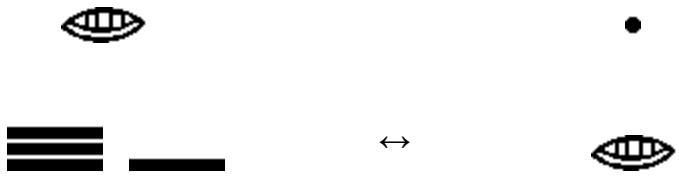
of numbers. Their place value numerical system with a zero did possess some advantages. They were able to represent infinitely large numbers by merely placing the appropriate amount of each appropriate symbol in each level. This eliminated the need for the invention of new symbols for infinitely large numbers and so the Mayas only had to utilize a very small quantity of symbols to read and write numbers.

**Arithmetic using the Mayan Numerals**

Even though there is no official evidence as to how the Mayas actually performed mathematical calculations using these numbers, it is possible to do arithmetic with these numbers including addition, subtraction, and multiplication. We will now discuss algorithms for Mayan addition, subtraction, and multiplication as discussed in Anderson's *Arithmetic in Mayan Numerals* partly based on my interpretation of Mayan arithmetic.

Below is an addition/subtraction table:

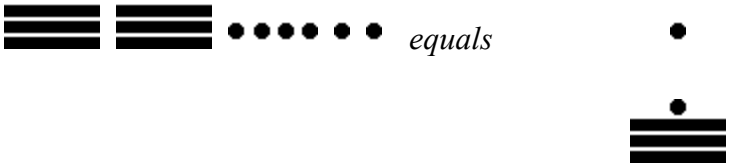
(1) 

(2) 

To perform addition, starting from the bottom level and working our way up, we find the sum of the total number of bars in the bottom level of both numbers and place that many bars in the bottom level. Similarly, we find the sum of the total number of dots in the bottom level of both

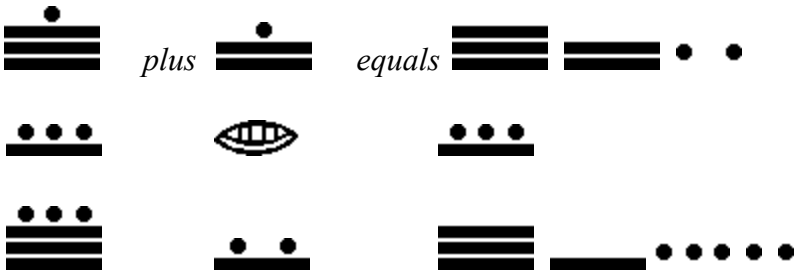
numbers and place that many bars in the bottom level. If there is a zero, we just ignore it since it is just a placeholder representing zero units. Then we follow the same procedure for the second level (the next level up), placing the dots and bars in the second level. Then we continue the procedure for all the other levels. Once we have done that, we apply the left to right direction of (1) and (2) in the addition/subtraction table to arrive at our final answer. Below are a couple of addition problems:

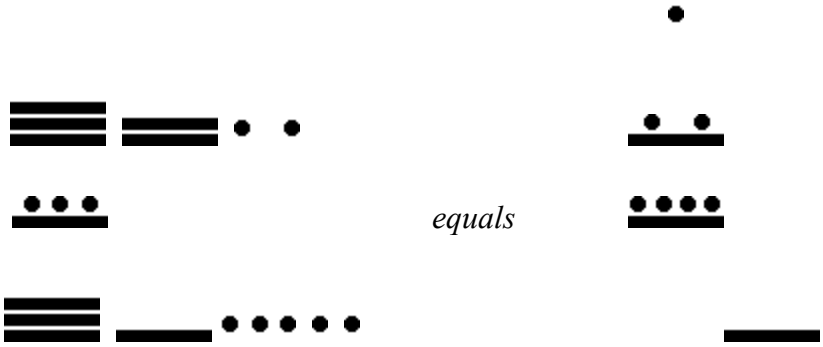
(1)


(2) 


In step (1), we found sum of the total number of bars in the bottom level of both numbers and placed that many bars in the bottom level. We followed the same procedure for dots. In step (2), we applied the left to right direction of (1) and (2) in the addition/subtraction table to arrive at our final answer.

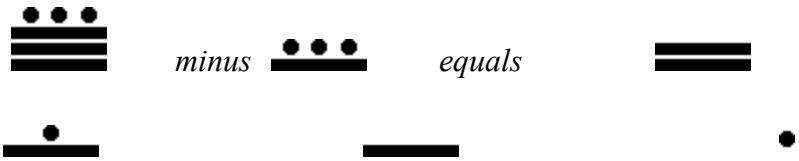
Similarly, the below addition problem proceeds as given below:

(1) 

(2) 


To perform subtraction, working from the bottom level and working your way up, we have to find the difference of the total number of bars in the bottom level of the bigger number and the total number of bars in the bottom level of the smaller number and place that many bars in the bottom level. Similarly, we have to find the difference of the total number of dots in the bigger number and the total number of dots in the smaller number and place that many dots in the bottom level. If the number of bars/dots) in the same level for the smaller level exceed the amount of bars/dots in the same level for the larger number, then apply the right to left direction of (1) and/or (2) in the addition/subtraction table. Since zero is merely a placeholder indicating the absence of units, we can also ignore the zero. Then we do the same procedure for the second level, placing the dots and bars in the second level. Then, we do the same procedure for all the other levels. Once we have done so, we will arrive at our final answer.

So the subtraction problem below should proceed as given,

(1) 


Similarly, this subtraction problem proceeds as given,

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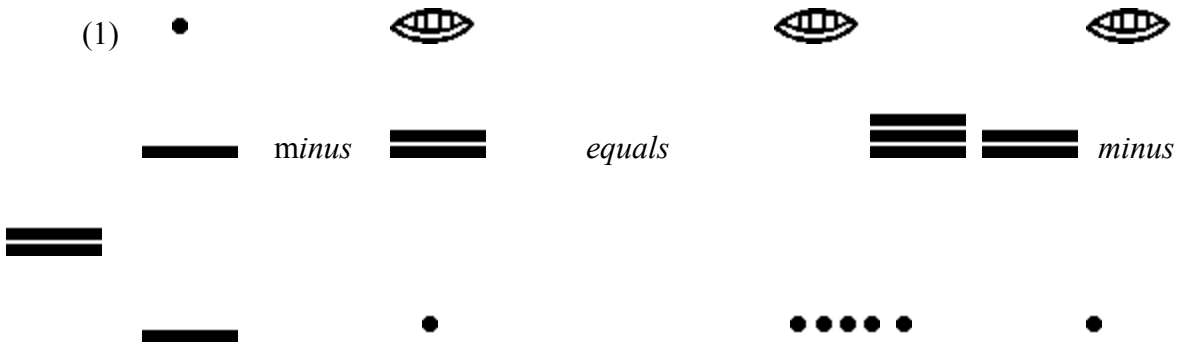
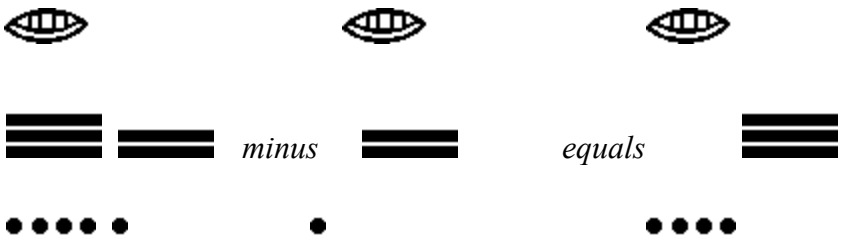
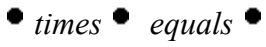
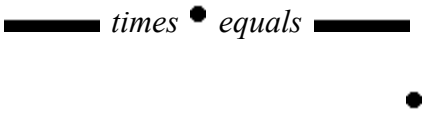
(1) 


(2) 

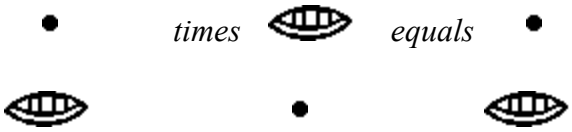

In step (1), the right to left direction of (1) and (2) in the addition/subtraction table was applied.

The table below is a multiplication table,

(1) 

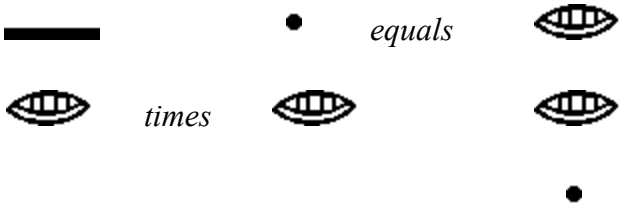

(2) 

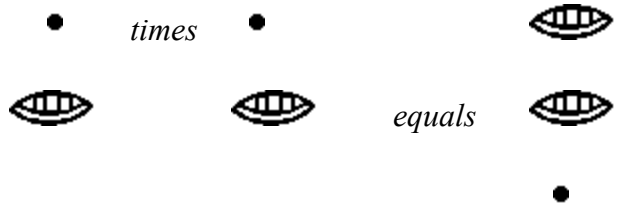

(3) 


(4) 

(5)
times
equals

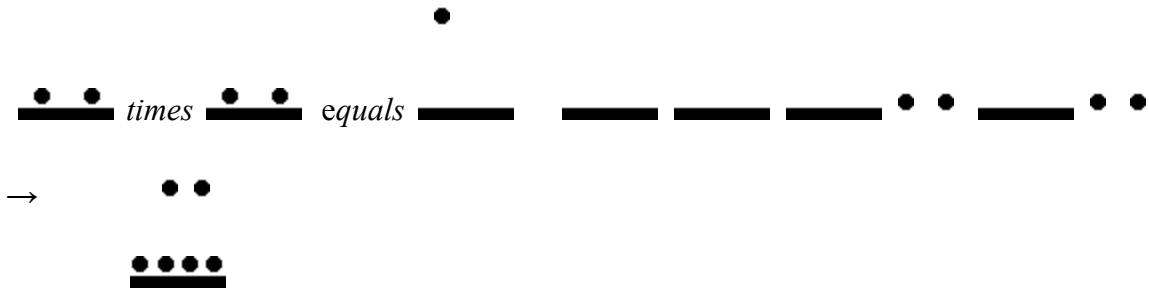
(6)
times
equals

(7) 
equals
times

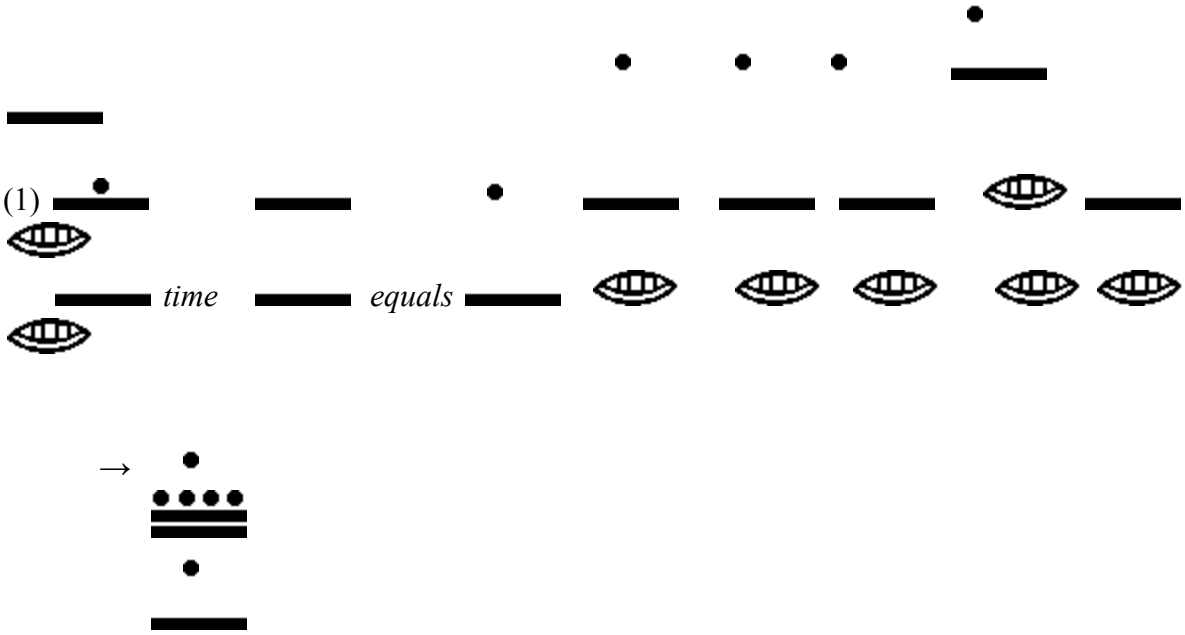
(8) 
times
equals

(9)
equals
times

Multiplication is commutative which means that the order of the numbers have no affect on the product. The same procedure applies for larger numbers and similar problems. Any number multiplied by zero is zero. We only need to apply the multiplication table to perform multiplication. The below multiplication problem proceeds as follows,

(1) 


(1), (2), and (3) was applied from the multiplication table and the number placed to the right were added to arrive at the final answer. It is important to take into account that just like in our number system, Mayan multiplication is a cumulative operation. In other words, the order of the multiplicand and the multiplier has no effect on the product. Similarly, the below multiplication problem is performed as given:

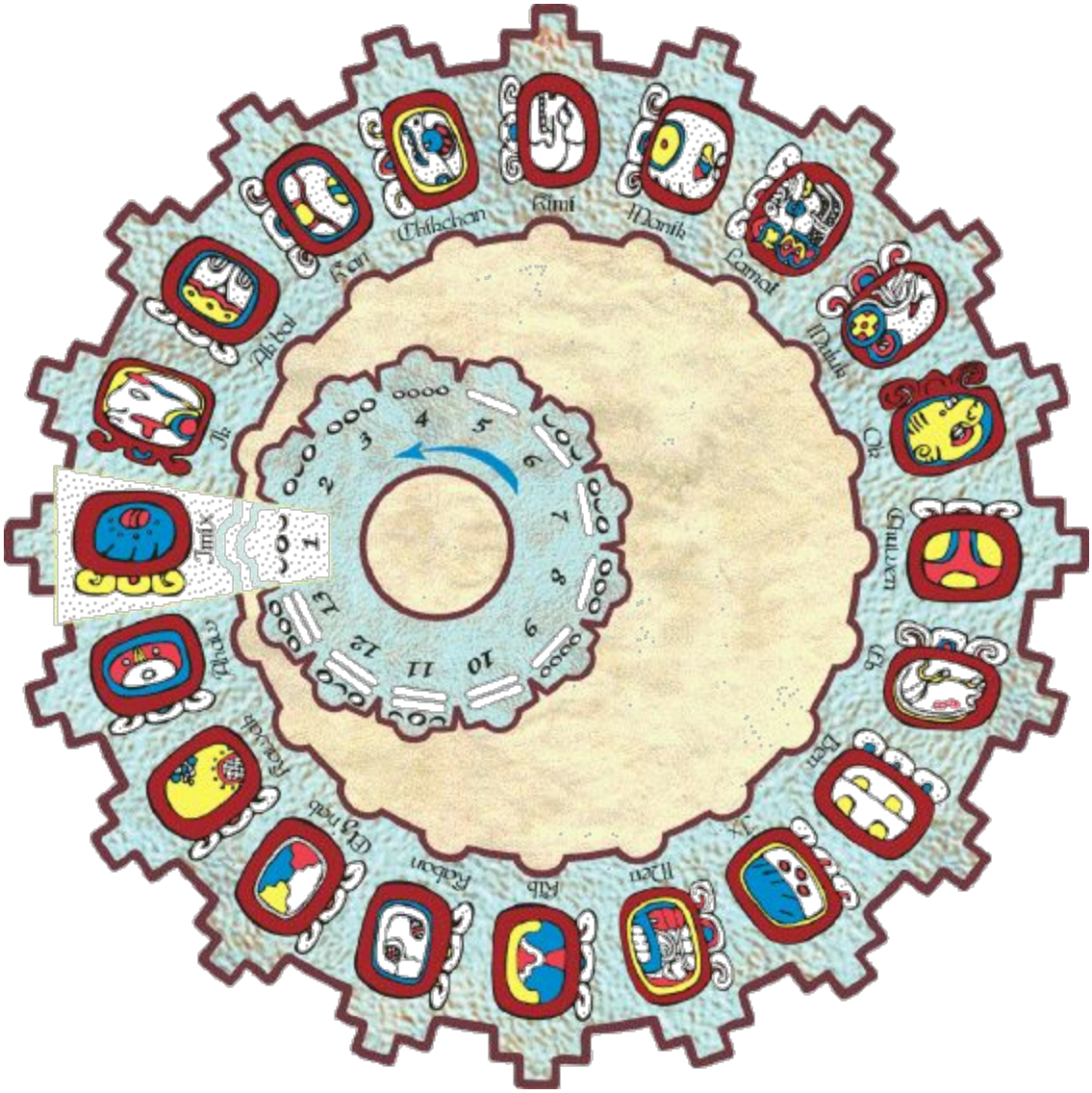
(1) 

In the multiplication problem, we applied (3), (6), (7), and (8) to arrive at the final answer. Once again no one knows how the Mayas did these mathematical calculations. All we know is that somehow they did perform these calculations.

**The Mayan Calendar**

As previously discussed, the Mayan calendar was developed during the Classic era. No one knows how the Mayas actually did these problems, but somehow the Mayas were able to solve these mathematical problems. According to Fuentes, the Mayas needed a way of keeping track of time so that they know when to plant corn, harvest corn, and clear the land, which was their main crop. The Mayas realized "between the sawing of corn and clearing it by burning 260 days lapsed", and therefore constructed this calendar based on the cycle of cultivation of corn.(SITE SOURCE). Therefore they constructed a calendar of 260 days called the Sacred Calendar (the tzolkín) that took track of the planting and harvesting days.

Below is a picture representing the tzolkín from the website www.armageddononline.org.




**PICTURE 1.**

They also needed a way of keeping track of the cycle of the sun since the year is divided into seasons in which there may be a rainy season versus a dry season or a hot season versus a cooler season. In addition, different seasons of the year have more sunlight than others, which is

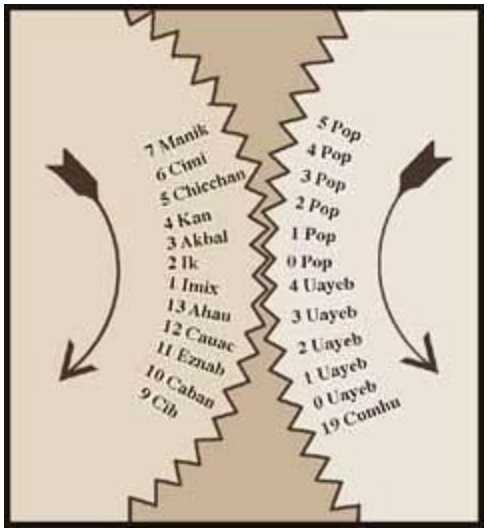
also important in the functionality of a complex society such as the Mayas. If they did not keep track of seasons, they would not have known when the next season is going to come and could have had a great impact on their agriculture. Therefore, they must know on what particular day in the solar cycle is each day of the Sacred Calendar. Therefore, since they almost correctly calculated that the sun takes 365 days to revolve around the sun, they created a calendar of 365 days that was based on the cycle of the sun around the earth called the haab (the Solar Year). Below is a table and picture of the haab from the websites http://www.acoustics.org/press/144th/monthsm.jpg, http://dosmanos.com/Merchant2/graphics/00000001/RAMACAL.jpg) respectively .

**Picture 2.**

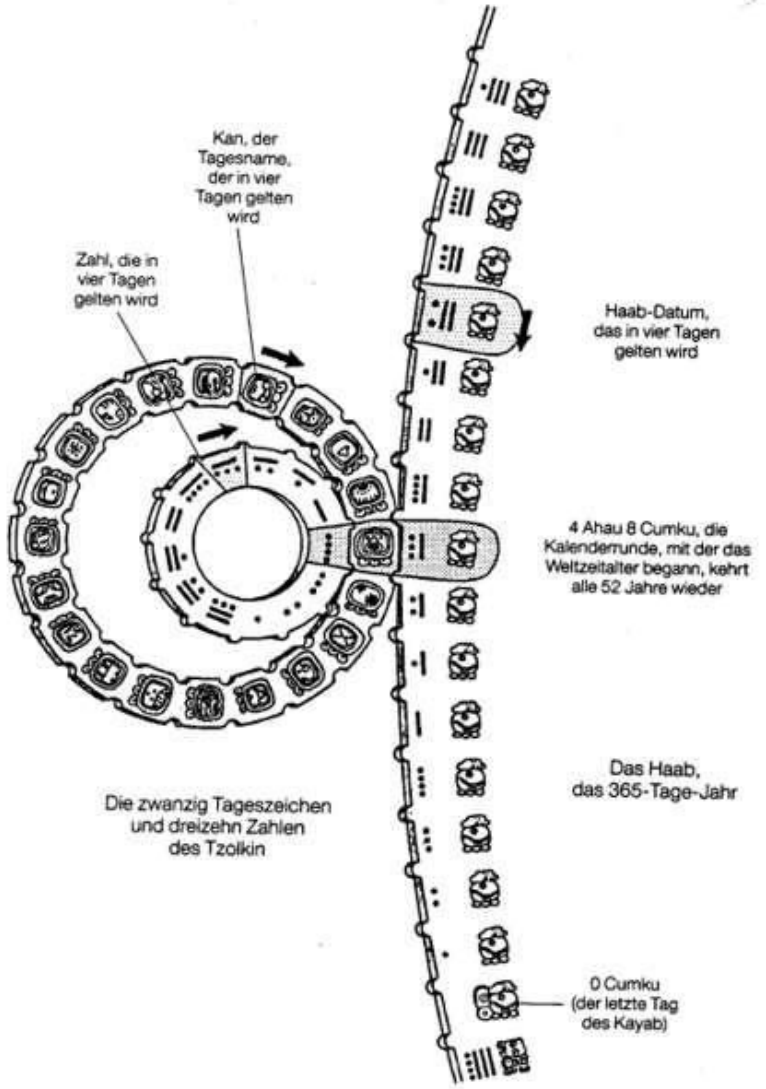
**Picture 4**.

Since both cycles of time were equally important for the Mayas, they used both calendars interchangeably and therefore a Mayan day consisted of a day in the Sacred Calendar followed by a day in the Solar Calendar. The calendar composed of both the Sacred Calendar and the Solar Calendar is called the Calendar Round. The set of all Calendar Round days is denoted R and will be discussed in more detail later. Even though it seems like there will be 365(260) = 94, 900 days in the Calendar Round, there are in far less days than that in the Calendar Round. Due to the mechanics of the Calendar Round, not all of these 94, 900 possible days are valid days for the Calendar Round. Since the Sacred Calendar has 260 days, the Solar Calendar has 365 days, and gcd(260, 365) =1, it follows the days in the Calendar Round do not

repeat until lcm(260, 365) = 18, 980 days later. Therefore, the Calendar Round is composed of only 18, 980 days. The Calendar Round corresponds to the days and months used in our Gregorian Calendar system. Even though many English sources written the numbers in decimal form and written the named days and months using the modern English alphabet, the Mayans always written the numbers in Mayan numeral form and used glyphs to represent the named days and months. Below are pictures of the Calendar Round.




Picture 3 (The modern English version of the Calendar Round)




Picture 4(The Mayan version of the Calendar Round)

The Sacred Calendar consisted of a cycle of thirteen days and a separate cycle of twenty named days in which the thirteen days were numbered from 1 to 13 and the twenty days were numbered from 1 to 20. The Mayans wrote the thirteen numbers in Mayan numerical form and the twenty days were numbered glyphs. Pictures of these glyphs and the names of these glyphs will be shown later on in this thesis, but for right now let's just focus on the Mathematics of the calendar. Let's define the set of all Mayan Sacred Calendar days as T={{1, 2, …., 13}×{1, 2,

…., 20}}. Even though some sources number the days from 0 to 19, in this context we are going to number them from 1 to 20. Therefore, starting from the first day which is (1, 1), a typical Sacred Year would proceed as follows:

(1, 1), (2, 2), (3, 3), (4, 4), . . .,(13, 13), (1, 14), (2, 15), (3, 16), . . .,(7, 20), (8, 1), (9, 2). The Solar Calendar on the other hand consisted of nineteen months in which eighteen months consisted of twenty days and the nineteenth month consisted of five days called the unlucky days. Let's define the set of all Mayan Solar Calendar days as $H=\{\{\{1, 2, \ldots, 20\}\times\{1, 2, \ldots, 18\}\} \cup \{\{1, \ldots, 5\} \times\{19\}\}\}$. Therefore starting from the first day of the year which is (1, 1), a typical Solar Year would go as follows:

(1, 1), (2, 1), (3, 1),…, (20, 1), (1, 2), (2,2), (3, 2),….(20, 18), (1, 19), (2, 19),…(5, 19) and then afterwards the new Solar Year starts again at (1, 1). Therefore since the Calendar Round is composed of both Calendars simultaneously, it follows a typical Calendar Round Year would go as follows:

((1, 1), (1,1)), ((2, 2), (2, 1)), …., ((13, 13), (13, 1)), ((1, 14), (14, 1)),…., ((7, 20), (20, 1)), ((8, 1), (1, 2)), ((9, 2), (2, 2)), …, ((13, 6), (6, 2)), ((1, 7), (7, 2)), …, ((9, 20), (20, 18)), ((10, 1), (1, 19)), ((11, 2),(2, 19)), …, ((1, 5), (5, 19)), ((2, 6), (1, 1)), …,((13, 20), (20, 18)) and the Calendar Round would begin again at ((1, 1), (1, 1)). A more detailed discussion of the set of all days of the Calendar Round will be discussed later.

We are going to discuss the arithmetic associated with the Calendar Round. The arithmetic consists of adding days to a Calendar Round day, subtracting kins from a Calendar Round day, and finding the number of days between two Calendar Round days. It is much more complicated to find the number of days between two Calendar Round days due to the complexity

of the Calendar Round and so for now we are going to discuss how to add and subtract Calendar Round days. To add N days to a Calendar Round day ((w, x), (y, z)), we have to add N days to the Sacred Calendar day and N days to the Solar Calendar day. Since the first portion of the Sacred Calendar consists of cycles of 13 days and the second portion of the Sacred Calendar consists of cycles of 20 days, to add N days to the Sacred Calendar day, we have to find (w′, x′) ε $Z_{13} \times Z_{20}$ such that w′ = (w+N)mod 13 and x′ = (x+N)mod 20. If w′ or x′ is equal to 0, then x′ is day number thirteen in the cycle and w′ is day number twenty in the cycle, which would mean (x′, w′) is N days after (x, w) in the Sacred Calendar. Since the Solar Calendar consists of eighteen months of twenty days with an extra month of five days, to add N days to the Solar Calendar day, we have to first see what day it would be in the Solar Calendar, add N days to the result, and then convert it back to day form. Therefore to add N days to (y, z), we have to find M where (y, z) is the Mth day of the Solar Year. To do that, we have to find M, such that M = (z-1)20+y. Now we have to find out what day in the Solar Year is N days after the Mth day of the Solar Year. To do that, we have to find a such that a = (M +N) mod 365 = ((z-1)20+y+N) mod 365. Now we have to represent the new day in Calendar form. If $1 \leq a \leq 360$, then we have to find z′ such that z′ =a mod 20 to find out what day of a month it would be. If $361 \leq a \leq 365$, then we have to find z′ such that z′ = a mod 5. To find out what month it would be we have to find y′ such that y′ = (a-z′+20)/20. Therefore, (y′, z′) is N days after (y, z) in the Solar Calendar. Therefore it follows that ((w′, x′), (y′, z′)) is N days after ((w, x), (y, z)). Whence in summary, for Calendar Round days ((w, x), (y, z)) and ((w′, x′), (y′, z′)),

((w′, x′), (y′, z′)) is N days after ((w, x), (y, z)) if and only if

w′ = (w+N) mod 13

$x' = (x+N) \bmod 20$

$y' = a \bmod 20$ where $(y', z')$ is the ath day of the Solar Year

$z' = (a-y'+20)/20$.

This summary leads us to a lemma:

**Lemma 1:**

Suppose $((w, x), (y, z))$ and $((w', x'), (y', z'))$ are Calendar Round days. Then $((w', x'), (y', z'))$ is N days after $((w, x), (y, z))$ if and only if

$w' = (w+N) \bmod 13$

$x' = (x+N) \bmod 20$

$y' = a \bmod 20$ or $y'=a \bmod 5$ where $(y', z')$ is the ath day of the Solar Year

$z' = (a-y'+20)/20$.

Use of this lemma provides a convenient way to add days. Also note that whenever 0 appears as a day in calculations, it should be replaced by 13 or 20 when converting to calendar notation.

Now we are going to discuss how to subtract kins from a Calendar Round day. Since subtraction is the inverse of addition, we can easily provide the algorithm and lemma for doing this operation from the process and lemma used for addition. So let $((w, x), (y, z))$ be a Calendar Round day. To find out what day in the Sacred Year is N days before $(w, x)$, we have to find $w'$, $x'$ where $w' = (w-N) \bmod 13$ and $x' = (x-N) \bmod 20$. Note that $1 \leq w' \leq 13$ and $1 \leq x' \leq 20$, therefore if we were to end up with a negative number, then the final answer would be the smallest positive number congruent to the negative number. To find out what day in the Solar Year is N days before $(y, z)$, we have to find M such that $(y, z)$ is the Mth day of the Solar Year. To do that, we find M such that $M = 20(z-1) + y$. Then, to find out what position the day that is

N days before (y, z), denoted (y′, z′), holds in the Solar Year, we have to find a such that a = (M-N) mod 365. Then to convert to calendar form, we have to find (y′, z′) such that y′ = a mod 20 and z′ = (a-z′+20)/20. Therefore ((w′, x′), (y′, z′)) is the Calendar Round day that is N days before ((w, x), (y, z)), and so we arrive at our second lemma:

**Lemma 2:**

Suppose ((w, x), (y, z)) and ((w′, x′), (y′, z′)) are Calendar Round days. Then ((w′, x′), (y′, z′)) is N days before ((w, x), (y, z)) if and only if

w′ = (w-N) mod 13

x′ = (x-N) mod 20

y′ = a mod 20 or y′ = a mod 5 where (y′, z′) is the ath day of the Solar Year

z′ = (a-y′+20)/20.

This lemma provides a convenient method of subtracting kins from a Calendar Round day. Also note that whenever 0 appears as a day in calculations, it should be replaced by 13 or 20 when converting to calendar notation.

Before discussing how to find the number of days between two Calendar Round dates, we first need a way of determining what days are Calendar Round days and what days are not since it will be impossible to determine the number of days between days that do not exist. We know that since the Solar Calendar is composed of eighteen months of twenty days and a nineteen month period with five days, it follows an arbitrary day

(y, z) of the Solar Year is the s+365p day of the Calendar Round where s, p are integers, where p is an integer such that $0 \leq p \leq 51$, where $1 \leq s \leq 365$, and where s = (z-1)20 + y. We also know an arbitrary day (w, x) of the Sacred Year is the 260q + r day of the Calendar Round, where r, q are

integers, where r is the rth day in the Sacred Calendar, $1 \leq r \leq 260$, and where $0 \leq q \leq 72$. From the composition of the Calendar Round, it follows ((w, x), (y, z)) is a Calendar Round day if and only if there exist an M such that $M = s + 365p = (z-1)20 + y + 365p$ and $M = 260q + r$. This further corresponds to $M \equiv r \bmod 260$

$$M \equiv ((z-1)20 + y) \bmod 365.$$

To further arrive at a solution, we can further break it down to

$$M \equiv r \bmod 13$$

$$M \equiv r \bmod 20$$

$$M \equiv ((z-1)20 + y) \bmod 365.$$

Since r is the rth day of the Sacred Calendar, it follows r is the rth day of the 13 day cycle and r is the rth day of the 20 day cycle. Therefore, $w \equiv r \bmod 13$ and $x \equiv r \bmod 20$ which means from the transitivity of congruence, $M \equiv w \bmod 13$ and $M \equiv x \bmod 20$. Therefore, it follows

$$M \equiv w \bmod 13$$

$$M \equiv x \bmod 20$$

$$M \equiv ((z-1)20 + y) \bmod 365.$$

We can apply the Chinese Remainder theorem to $M \equiv w \bmod 13$

$$M \equiv x \bmod 20$$

to solve the congruence in terms of w and x. To find the M that satisfies the equation, we need to find n such that $n = 13(20) = 260$. We also need to find $N_1$ and $N_2$ such that $N_1 = 260/13 = 20$ and $N_2 = 260/20 = 13$. Then we must find $x_1$ and $x_2$ such that

$20x_1 \equiv 1 \bmod 13$ and $13x_2 \equiv 1 \bmod 20$. It turns out that $x_1 = 2$ and $x_2 = 17$. From here we find X such that $X = 2(20)w + 17(13)x \bmod 260 = 40w + 221x \equiv r \bmod 260$. Therefore from here, it follows ((w, x), (y, z)) is a Calendar Round day if and only if

$M \equiv r \bmod 260$

$M \equiv ((z-1)20 + y) \bmod 365$

if and only if $M \equiv 40w+221x \bmod 260$

$M \equiv ((z-1)20+y) \bmod 365$

if and only if $40w+221x+260q=(z-1)(20)+y+365p$

if and only if $260q-365p = (z-1)(20)+y-40w-221x$

if and only if $260q-365r = (z-1)20 + y - 40w-221x$

if and only if $260q-365r = (z-1)20 + y - 40w-221x$ has a solution for q and r

if and only if $5|(z-1)20 + y - 40w-221x$

if and only if $5|y-221x$

if and only if $y \equiv 221x \bmod 5$

if and only if $y \equiv x \bmod 5$.

Therefore we arrive at our third lemma:

**Lemma 3:**

Suppose $(w, x) \in T$ and $(y, z) \in H$. Then $((w, x), (y, z)) \in R$ if and only if $y \equiv x \bmod 5$.

Therefore the set of all Calendar Round days is defined as $R = \{T \times H \mid (w, x) \in T \text{ and } (y, z) \in H \text{ and } y \equiv x \bmod 5\}$.

To give an example, $((3, 4), (9, 5)) \in R$ because $4 \equiv 9 \bmod 5$ and

$((9, 7), (4, 6)) \notin R$ because 7 is not congruent to 4 modulo 5.

Now that we have a way of telling when two Calendar Round days actually exists, we can proceed to find the number of days between two Calendar Round days. So let $((w, x), (y, z)), ((w', x'), (y', z')) \varepsilon R$ where $((w', x'), (y', z'))$ occurs after $((w, x), (y, z))$. We need to find an N such that

$$N \equiv (w'-w) \bmod 13$$

$$N \equiv (x'-x) \bmod 20$$

$$N \equiv ((z'-1)20 + y')-((z-1)20+y)) \bmod 365$$

$$\equiv 20((z'-1)-(z-1))+( y'-y) \bmod 365.$$

Since gcd(20, 365) >1, it follows we cannot use the Chinese Remainder theorem until we find a way to rewrite this linear congruence. We can further simplify this to

$$N \equiv (w'-w) \bmod 13$$

$$N \equiv (x'-x) \bmod 5$$

$$N \equiv (x'-x) \bmod 4$$

$$N \equiv 20((z'-1)-(z-1))+( y'-y) \bmod 5$$

$$N \equiv 20((z'-1)-(z-1))+( y'-y) \bmod 73$$

which can be simplified to

$$N \equiv (w'-w) \bmod 13$$

$$N \equiv (x'-x) \bmod 5$$

$$N \equiv (x'-x) \bmod 4$$

$$N \equiv ( y'-y) \bmod 5$$

$$N \equiv 20((z'-1)-(z-1))+( y'-y) \bmod 73.$$

From here since $y \equiv x \bmod 5$ and $y' \equiv x' \bmod 5$, it follows

$N \equiv (x'-x) \equiv (y'-y) \bmod 5$ which means if $N \equiv (x'-x) \bmod 5$, then $N \equiv (y'-y) \bmod 5$.

Therefore it follows that we can further simplify this problem to:

$$N \equiv (w'-w) \bmod 13$$

$$N \equiv (x'-x) \bmod 5$$

$$N \equiv (x'-x) \bmod 4$$

$$N \equiv 20((z'-1)-(z-1))+(y'-y) \bmod 73$$

Now that greatest common divisor of all the moduluses is 1, we can apply the Chinese Remainer Theorem. We need to find n such that $n = 13(5)(4)(73) = 18980$. Then we must find N1 such that $N1 = 18,980/13 = 1460$, $N2 = 18980/5 = 3796$, $N3 = 18980/4 = 4745$, and $N4 = 18980/73 = 260$. Then we must find x1, x2, x3, and x4 such that $1460x1 \equiv 1 \bmod 13$, $3796x2 \equiv 1 \bmod 5$, $4745x3 \equiv 1 \bmod 4$, and $260x4 \equiv 1 \bmod 73$. It turns out $x1 = 10$, $x2 = 1$, $x3 = 1$, and $x4 = 57$. Therefore it follows we produce X such that $X = 10(1460)(w' - w) + 1(3796)(x' - x) + 1(4745)(x' - x) + 57(260)(20((z'-1)-(z-1))+(y'-y)) \bmod 18980$. Now we want to find the smallest positive integer N such that $N \equiv X \equiv 10(1460)(w' - w) + 1(3796)(x' - x) + 1(4745)(x' - x) + 57(260)(20((z'-1)-(z-1))+(y'-y)) \bmod 18980$. This can be further simplified to $N \equiv X \equiv (14,600)(w' - w) + (8541)(x' - x) + (14,820)(20((z'-1)-(z-1))+(y'-y)) \bmod 18980$. Thus we arrive at a new lemma.

**Lemma 4:**

Let $((w, x), (y, z)), ((w', x'), (y', z')) \varepsilon R$ where $((w, x), (y, z))$ occurs before the latter. Then, there are N days between $((w, x), (y, z))$ and $((w', x'), (y', z'))$ and N is given by the smallest positive integer such that

$N \equiv (14{,}600)(w' - w) + (8541)(x' - x) + (14{,}820)(20((z'-1)-(z-1))+(y'-y)) \bmod 18980.$

To give one example, for the Calendar Round dates ((1, 2), (2, 6)) and ((3, 4), (9, 9)), applying the theorem, we have $N \equiv 14{,}600(2) + (8541)(2) + (14{,}820)(20(8-5)+ 7)$

$$\equiv 29{,}200 + 17082 + 992{,}940$$

$$\equiv 1{,}039{,}222$$

$$\equiv 8{,}079 \bmod 18980$$

which means there are 8, 079 days between the two dates.

The Mayas in addition to the Calendar Round utilized the Long Count system in their calendar system. The Long Count measure the number of days elapsed since a starting point which is the beginning of the current Great Cycle to arrive at the current Calendar Round date, which is a period of about 5, 128 Solar Years. The last Great Cycle ended on August 13, 3114 B.C. on the Gregorian Calendar and the current Great Cycle will end on December 21, 2012. It was important for them to know this information so they can keep track of important historical events such as when important leaders died. If there was no way of recording the amount of time passed since a fixed point, then it would be very difficult to keep track of sequences of important events and when exactly did they occur. It would be the same as if we were to just say Julius Caesar died on January 5. We would have no idea whether this event occurred recently, during the renaissance, during the imperial age, or during the Roman Empire era. In addition, if an event happened on December 20 and another event happened on July 7, it would be equally likely that both occurred before Caesar died, both occurred after Caesar died, or either one occurred before Caesar and the other died after Caesar. Therefore we would have no idea who died first. Thus it is insufficient to just record the Calendar Round days when recording a date,

and so the Long Count must be included to record a complete date. The Long Count is very similar to the vigesimal numerical system that we discussed with just one modification. Instead of $20^0$, $20^1$, $20^2$, $20^3$, etc. as in the vigesimal number system, in calendar notation, we have $20^0$, $20^1$, $20^1(18)$, $20^2(18)$, $20^3(18)$, etc. This change is implemented so that the Long Count can closely approximate the Solar Year, for 360 days is much closer to 365 days than 400 days. The units of the Long Count are known as kins (or days) and the place value of the Long Count goes as described below:

$20^0$ = 1 day = 1 kins

$20^1(1)$ = 20 kins = 1 uinal

$20^1(18)$ = 360 kins = 1 tun

$20^2(18)$ = 7200 kins = 1 katun

$20^3(18)$ = 144, 000 kins = 1 baktun

$20^4(18)$ = 2, 880, 000 kins = 1 pictun

$20^5(18)$ = 57, 600, 000 kins = 1 calabtun

$20^6(18)$ = 1, 152, 000, 000 kins = 1 kinchiltun

$20^7(18)$ = 23, 040, 000, 000 days = 1 alautun.

According to Robert J. Sharer and Loa P. Traxler's *The Ancient Maya,* "although these terms are used by scholars for convenience, the ancient Maya names for these time units are only partially known" (SITE SOURCE). The Great Cycle is equivalent to 13 baktuns (1, 872, 000 days), therefore pictuns, calabtuns, kinchiltuns, and alautuns are not used in Long Count dates. However, they can be used to date information from the distant past as far back as billions of years. According to Ascher's *Mathematics Elsewhere,* there was an inscription of the Calendar

Round dates of the birth and enthronement of a ruler that occurred 7 calabtuns, 18 pictuns, 2 baktuns, 9 katuns, 2 tuns, 12 uinals, and 1 days (1. 25 million years) before it was inscribed.

We denote the set of all Long Count dates L as

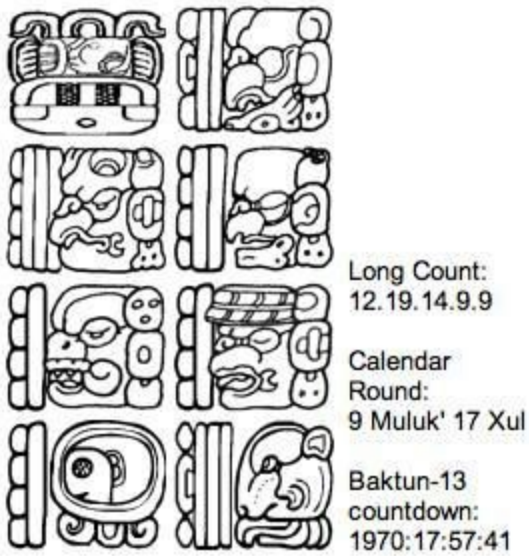
$L = \{0, 1, 2,\ldots, 19\} \times \{0, 1, 2, \ldots., 17\} \times \{0, 1, 2, \ldots., 19\} \times \{0, 1, 2, \ldots, 19\}$

$\times \{0, 1, 2, \ldots. , 19\}$.

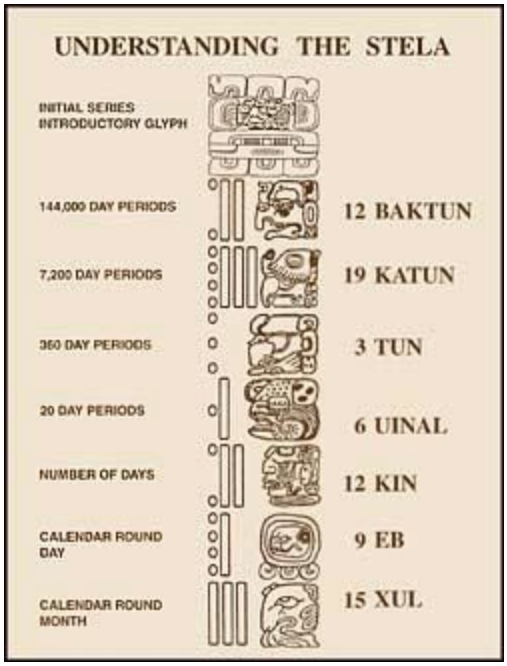
Therefore a Long Count date is of the form (a, b, c, d, e) where (a, b, c, d, e) ε L. It can also be written as a. b. c. d. e. and most English texts write Long Count dates in this form. However, the Mayans wrote Long Count dates using the same glyphs as they used for numbers in their appropriate place values and glyphs for the different units of time. The following pictures demonstrate more the characteristics of the Long Count. As mentioned earlier, the simultaneous use of the Calendar Round and the Long Count system corresponds to our system of days, months, and years in the sense that a date is not complete in our system if we do not include the date, month, and year. Therefore, it is important to remember that the combination of the Long Count date and the Calendar Round date is just as important in their calendar system as the combination of days, months, and years in our Gregorian Calendar. So, the Mayan date 4 Imix 9 Pop 1. 4. 5. 8. 9. means 1 baktun, 4 katuns, 5, tuns, 8 uinals, and 9 kins elapsed since the start of the current Great Cycle to arrive at the current Calendar Round date of 4 Imix 9 Pop. Below are other examples of Long Count glyphs and complete Mayan date using the Calendar Round and Long Count from http://members.shaw.ca/mjfinley/LCgeo.gif, from http://raphael.doxos.com/wp-content/uploads/2007/07/maya20070730.jpg, from http://www.mayacalendar.com/imagen/table.jpg respectively:

| | GEOMETRIC | CODEX |
|---|---|---|
| PIK'TUN (20 BAK'TUNOB) | | |
| BAK'TUN (20 K'ATUNOB) | | |
| K'ATUN (20 TUNOB) | | |
| TUN (18 WINALOB) | | |
| WINAL (20 K'INOB) | | |
| K'IN (1 DAY) | | |

**Table 2**. Long Count glyphs




**Picture 4.**: One version of a complete Mayan date



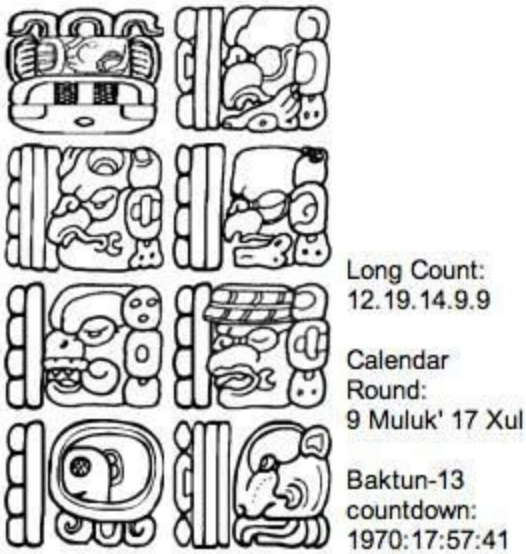

**Picture 5.** another complete Long Count date

Now that we have introduced the Long Count, we are going to discuss the arithmetic that can be used with the Long Count. The arithmetic that can be used with the Long Count system includes addition of a Long Count number of days to a Long Count date, the subtraction of a Long Count number of days to a Long Count date, and finding the Long count number of days between two Long Count dates. The algorithm for the arithmetic for the first two operations is identical to the algorithm for the arithmetic used for the regular vigesimal number system with one modification. There could be seventeen units at most in the unials place value. The latter algorithm is identical to the algorithm for subtraction of Long Count days from a Long Count date with one modification; the answer is a Long Count number of days and not a Long Count day.

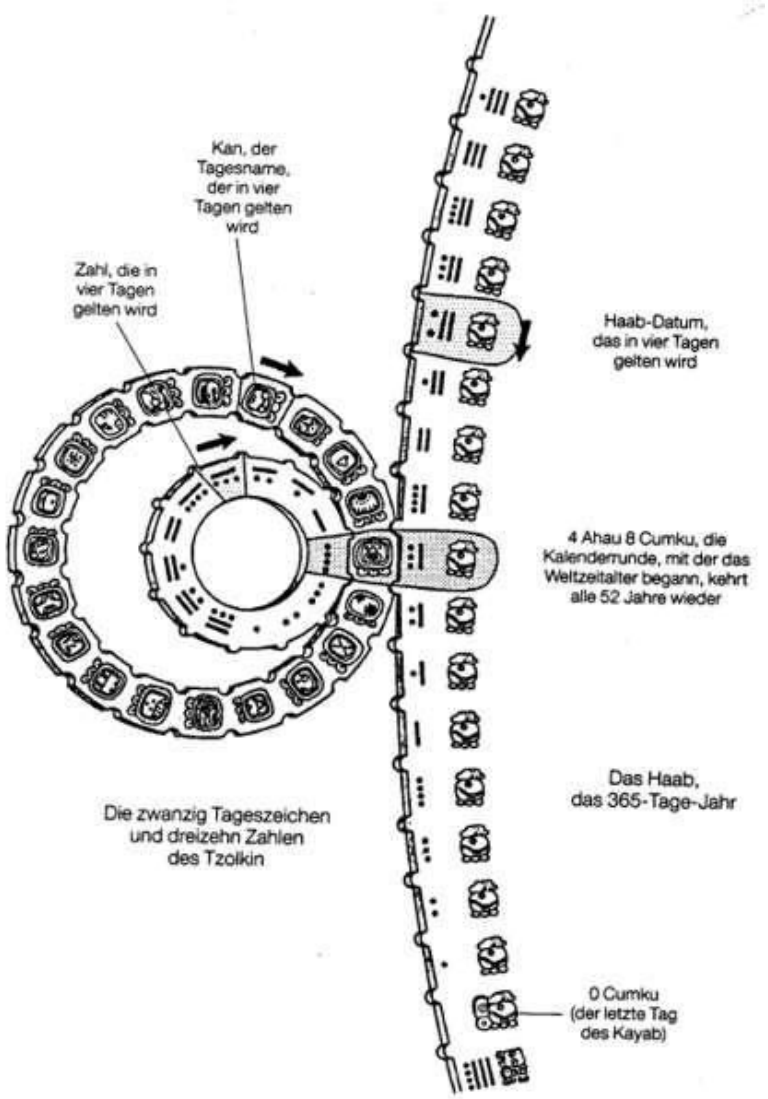
PICTURE 1: The Tzolkín from http://www.armageddononline.org/image/mayan1.GIF

PICTURE 2: Mayan writing on a wall from http://www.hermetic.ch/cal_stud/maya/cd5024.jpg

PICTURE 3: The Haab ( from
http://dosmanos.com/Merchant2/graphics/00000001/RAMACAL.jpg)



PICTURE 4: A complete Mayan date from
http://raphael.doxos.com/wp-content/uploads/2007/07/maya20070730.jpg




PICTURE 5: Calendar Round http://www.sven-gronemeyer.de/figures/schrift_funktioncr.jpg

Nowadays, descendants of the Mayas live in southern Mexico and in the highlands of Guatemala. The contemporary Mayas live on farms where they farm mostly beans, squash, and corn surrounding villages and communities which are mainly composed of public buildings. More than two million people speak the modern form of the Mayan language, practice Mayan

traditions, and some of the contemporary Mayas speak Spanish as well. Even though the contemporary Mayas are Roman Catholic, a major part of their religion is composed of Mayan cosmology.

Much of what has been known about the Mayan empire come from stone inscriptions called *stelae* where they recorded important dates and events, four religious books known as the *codices*, traditions and customs that are still practiced today by the descendants of the Mayas, written records in Spanish of a Spaniard Diego de Landa who destroyed written records of the Mayas, and the discovery of the ancient cities, pyramids and buildings that were buried in the jungles. However, much of the Mayan culture, texts, and artifacts have been destroyed by the Spaniards who had little if any tolerance for their culture and therefore there is limited available archeological evidence that can tell us about the modern Mayan empire. Much of what has been published has been derived from the limited archeological evidence as well as the descendants of the Mayas. More still has to be learned about the Mayan empire, after all no one still knows how the Mayas actually solved these mathematical problems and who did. In addition, there is still decipherments of the hieroglyphics that have to be made.

It turns out the Mayan empire was quite an advanced empire. In fact, it was far more advanced than most Europeans could have imagined. The Mayas created a calendar system that enabled them to record time and important events so that they can carry out their daily activities in addition to a number system that enabled them to count and quantify. All of this was created by the priests and the elites of the Mayan society who lived in cities and population centers, and ruled over the commoners. The Mayas had all of the characteristics of an advanced society, even though they did not have any horses, wheels, metal swords, or guns. However, the Europeans

were too egocentric to realize the potential of the Mayan society and just destroyed it. Now the only challenges facing the advancement of knowledge of the Mayan empire is the decipherment of hieroglyphics, the discovery of more buildings, written sources, and artifacts, finding out more information from the indigenous people, and deriving empirical conclusion based on what has been found.

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